THE DENSITY OF SUFFIX-REPLACEABLE PAIRS IN PENNEY'S ANTE

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Abstract. Drexel, Peng and Richey studied the pairs (v, w) of binary words of the same length for which, in Penney's ante, the final copy of either word can be replaced by the other without creating a further occurrence of v or w. For such pairs with $v \neq w$ the probability that v appears before w has a simple closed form. They conjectured that the proportion of such ordered pairs among all pairs of length n converges to a positive limit. We prove this conjecture. Using a finite criterion in terms of the prefixes determined by the overlaps of v and w, we show that, outside an event of small probability, the property depends only on the first and last K letters of the two words. Consequently the proportion converges, with error $O(n^2 2^{-n/3})$, and the proportion among pairs of distinct words has the same limit. Combining a boundary construction of Bóna, Maga and Richey with an exact calculation of the probabilities of long cross-overlaps, we obtain the explicit lower bound $1/512$ for the limit.

1. Introduction

Let v and w be finite binary words, and generate independent random bits until one of the two words appears as a block of consecutive bits. This race is known as Penney's ante. Conway found a formula for the probability that v appears first; proofs and generalizations are due to Li [6] and to Guibas and Odlyzko [4]. Drexel, Peng and Richey [2] studied the win probability as a function of the bias p of the bits, in particular its symmetries under $p \mapsto 1 - p$ and the bijections that explain them. The following class of pairs plays a central role in their work.

For binary words v and w, let $\Omega^*_{v,w}(v)$ be the set of finite binary words u, possibly empty, such that v occurs in uv exactly once, namely as its suffix, and w does not occur in uv. The set $\Omega^*_{v,w}(w)$ is defined by interchanging v and w [2, Definition 2]. A pair (v, w) of words of the same length has *property R* if

$$\Omega^*_{v,w}(v) = \Omega^*_{v,w}(w). \tag{1.1}$$

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In other words, a game won by v becomes a game won by w when the final copy of v is replaced by w, and conversely; we call such pairs *suffix-replaceable*. If (v, w) has property R and $v \neq w$, then the probability that v appears before w under independent Bernoulli(p) bits, $0 < p < 1$, equals

$$\frac{\mathbb{P}_p(v)}{\mathbb{P}_p(v) + \mathbb{P}_p(w)}, \qquad \text{where } \mathbb{P}_p(x) = p^{|x|_1}(1-p)^{|x|_0}$$

and $|x|_a$ is the number of letters of x equal to a [2, Theorem 13]. Every pair of distinct words without nontrivial self-overlaps or cross-overlaps has property R, but there are many others [2, p. 8].

Let $\mathcal{R}_n$ be the set of ordered pairs $(v, w) \in \{0,1\}^n \times \{0,1\}^n$ with property R. Drexel, Peng and Richey conjectured [2, Conjecture 14] that $4^{-n}|\mathcal{R}_n|$ converges to a strictly positive limit. A positive lower bound for this proportion is implicit in the remark following Theorem 3.6 of Bóna, Maga and Richey [1, p. 10]: two words that begin with 00 and end with 11 have no short overlaps, the longer overlaps can be excluded by a union bound, and pairs without overlaps have property R [2, p. 8]. The existence of the limit is the main point of the present paper.

Theorem 1.1. *The limit*

$$\rho = \lim_{n\to\infty} 4^{-n}|\mathcal{R}_n|$$

exists, and $\rho \geq 1/512$. *Moreover, if* $\mathcal{R}_n^{\neq} = \{(v, w) \in \mathcal{R}_n : v \neq w\}$, *then*

$$\lim_{n\to\infty} \frac{|\mathcal{R}_n^{\neq}|}{4^n} = \lim_{n\to\infty} \frac{|\mathcal{R}_n^{\neq}|}{4^n - 2^n} = \rho.$$

Under the literal definition (1.1), every diagonal pair (v, v) belongs to $\mathcal{R}_n$, since both sides of (1.1) are empty (Remark 2.3). The second statement shows that the limit is unchanged if only pairs of distinct words, for which the race is non-trivial, are counted. Thus Theorem 1.1 proves [2, Conjecture 14] under either reading of the definition.

The proof gives a rate of convergence. For $K \geq 1$ let q_K be the probability that two independent uniformly random binary words of a common length $n > 2K$ satisfy condition (3.1) below; this condition involves only the first and last K letters of the two words, and q_K does not depend on n.

Theorem 1.2. *For every* $K \geq 1$ *we have* $|\rho - q_K| \leq 2^{2-K}$. *For every* $n \geq 3$, *with* $K = \lfloor n/3 \rfloor$,

$$\left|4^{-n}|\mathcal{R}_n| - \rho\right| \leq \big((K+1)(2K+1) + 8\big)\, 2^{-K} = O\big(n^2\, 2^{-n/3}\big).$$

Outline. The proof has three steps. First, property R is equivalent to a finite condition: two sets of proper prefixes of v and w are read off from the overlaps among v and w in all four directions, and every element of either set must have a suffix in the other set (Proposition 2.2). This is a reformulation of a criterion of Drexel, Peng and Richey [2, Lemma 19]. Second, for random words, with probability $1 - O(2^{-K} + K^2 2^{-(n-2K)})$ there is no overlap longer

than K, and none of the relevant prefixes is a suffix of another. On this event property R depends only on the first and last K letters of the two words (Lemma 3.3). The resulting estimate for $|4^{-n}|\mathcal{R}_n| - q_K|$ shows that the sequence is a Cauchy sequence. Third, for words that begin with 0^L and end with 1^L, we compute the conditional probability of each cross-overlap exactly, including overlaps of length close to n, where boundary letters of both words are matched against each other, and we bound the conditional probability of a self-overlap of length k by 2^{-k} (Lemma 4.1). For $L = 2$ this is the construction of [1], and it gives $\rho \geq 1/512$. This bound is far below the values of $p_n = 4^{-n}|\mathcal{R}_n|$ observed numerically in Table 1; see Remark 4.5.

Related work. Limiting densities of words and pairs of words defined by overlap conditions have been studied before. Holub and Shallit [5, Theorem 3] proved that the probability that a random word has a given maximal border length converges as the length grows. Gabric [3, Theorem 10 and Corollaries 11–12] proved the existence of limiting densities for pairs of words classified by the presence of overlaps in the two directions between them, and Rivals and Wang [7] counted the pairs of words with a prescribed correlation and obtained asymptotic bounds for the proportion of such pairs [7, Theorem 22]. Property R is not a condition on a single overlap set: by Proposition 2.2 it compares the overlaps in the four directions (v, v), (v, w), (w, v) and (w, w) through suffix relations between prefixes. It is not determined by the cross-overlaps, and it does not force the sets of self-overlap lengths of v and w to coincide (Example 2.4). To our knowledge, the existence of the limit in Theorem 1.1 has not appeared in the literature.

Organization. Section 2 fixes notation and proves the finite criterion. Section 3 proves the localization lemma and the estimate for $|4^{-n}|\mathcal{R}_n| - q_K|$. Section 4 treats words with prescribed boundary letters and gives the lower bound. Section 5 completes the proofs of Theorems 1.1 and 1.2.

2. Preliminaries and a finite criterion

All words are over the alphabet $\{0, 1\}$, and $\{0, 1\}^*$ is the set of finite words, including the empty word. For a word $a = a_1 \cdots a_n$ and $1 \leq i \leq j \leq n$ we write $a[i, j] = a_i \cdots a_j$, and for $0 \leq j \leq n$ we write $\mathrm{pre}_j(a) = a_1 \cdots a_j$ and $\mathrm{suf}_j(a) = a_{n-j+1} \cdots a_n$. A word y is a *suffix* of x if $x = zy$ for some word z; we allow $y = x$. For a set X of words, $\{0, 1\}^*X$ is the set of words having a suffix in X. A word a *occurs* in x if $a = x[i, i + |a| - 1]$ for some i; we then say that this occurrence *ends* at position $i + |a| - 1$.

Fix $n \geq 1$. For $s, t \in \{0, 1\}^n$ and $1 \leq k \leq n - 1$, we say that (s, t) has an *overlap of length* k if $\mathrm{suf}_k(s) = \mathrm{pre}_k(t)$. Following [2, Definition 6], put

$$D(s, t) = \{\mathrm{pre}_{n-k}(s) : 1 \leq k \leq n - 1,\ \mathrm{suf}_k(s) = \mathrm{pre}_k(t)\}. \tag{2.1}$$

For a pair (v, w) of words of length n we set

$$B_v = D(v, v) \cup D(w, v), \qquad B_w = D(v, w) \cup D(w, w). \tag{2.2}$$

Let $F_{v,w}$ be the set of words in which neither v nor w occurs. Every element of $B_v \cup B_w$ has length at most $n-1$ and therefore lies in $F_{v,w}$. When the pair is fixed we write $\Omega^*(v) = \Omega^*_{v,w}(v)$ and $\Omega^*(w) = \Omega^*_{v,w}(w)$.

Lemma 2.1. *Let $v, w \in \{0,1\}^n$ be distinct. Then*

$$\Omega^*(v) = F_{v,w} \setminus \{0,1\}^* B_v, \qquad \Omega^*(w) = F_{v,w} \setminus \{0,1\}^* B_w.$$

Proof. By symmetry it suffices to prove the first identity. Let $u \in \{0,1\}^*$. If $u \notin F_{v,w}$, then an occurrence of v or w in u is also an occurrence in uv. It ends before the last letter of uv, so it is a second occurrence of v or an occurrence of w, and $u \notin \Omega^*(v)$.

Let $u \in F_{v,w}$. The final n letters of uv form v, and they do not form w because $v \neq w$. Suppose that $a \in \{v, w\}$ has another occurrence in uv. It is not contained in u, and it ends before the last letter, so it ends at position $|u| + k$ with $1 \le k \le n-1$. Its last k letters are $\mathrm{pre}_k(v)$, and its first $n-k$ letters are $\mathrm{suf}_{n-k}(u)$. Hence $\mathrm{suf}_k(a) = \mathrm{pre}_k(v)$, and $\mathrm{pre}_{n-k}(a)$, an element of $D(a, v) \subseteq B_v$, is a suffix of u. Conversely, suppose that u has a suffix $\mathrm{pre}_{n-k}(a)$ with $a \in \{v, w\}$, $1 \le k \le n-1$ and $\mathrm{suf}_k(a) = \mathrm{pre}_k(v)$. Then

$$(uv)[\,|u| - n + k + 1,\ |u| + k\,] = \mathrm{pre}_{n-k}(a)\ \mathrm{pre}_k(v) = \mathrm{pre}_{n-k}(a)\ \mathrm{suf}_k(a) = a,$$

so a occurs in uv ending at position $|u| + k < |uv|$. This is either a second occurrence of v or an occurrence of w, and $u \notin \Omega^*(v)$. Therefore an element u of $F_{v,w}$ belongs to $\Omega^*(v)$ if and only if $u \notin \{0,1\}^* B_v$. □

Proposition 2.2. *Let $v, w \in \{0,1\}^n$ be distinct. Then (v, w) has property R if and only if*

$$\begin{gathered} \textit{every element of } B_v \textit{ has a suffix in } B_w, \textit{ and} \\ \textit{every element of } B_w \textit{ has a suffix in } B_v. \end{gathered} \tag{2.3}$$

Proof. If (2.3) holds, then each of the sets $\{0,1\}^* B_v$ and $\{0,1\}^* B_w$ contains the other, and Lemma 2.1 gives $\Omega^*(v) = \Omega^*(w)$. Conversely, suppose that $x \in B_v$ has no suffix in B_w. Then $x \in F_{v,w}$ since $|x| < n$, and $x \in \{0,1\}^* B_v$. By Lemma 2.1, $x \notin \Omega^*(v)$ and $x \in \Omega^*(w)$, so (1.1) fails. The case of an element of B_w without a suffix in B_v is symmetric. □

Remark 2.3. Drexel, Peng and Richey [2, Lemma 19] showed that property R can be decided by finitely many tests involving the set $D(v, v) \cup D(w, w) \cup D(v, w) \cup D(w, v)$. For distinct words, Proposition 2.2 is a reformulation of their criterion; we use the present form because it involves only suffix relations between elements of B_v and B_w. If $v = w$, then the final copy of v in uv is an occurrence of w, so $\Omega^*(v) = \Omega^*(w) = \emptyset$; hence every diagonal pair has property R, and

$$|\mathcal{R}_n| = |\mathcal{R}_n^{\neq}| + 2^n. \tag{2.4}$$

For $n = 1$ the sets B_v and B_w are empty, and all four pairs have property R.

Example 2.4. For $(v, w) = (00, 11)$ neither word overlaps the other, but $B_v = \{0\}$ and $B_w = \{1\}$, so the pair does not have property R. In particular, property R is not implied by the absence of cross-overlaps, and the density of Theorem 1.1 cannot be read off from the densities of pairs classified by cross-overlaps as in [3]. For the pair $(v, w) = (000100, 001110)$ of [2, p. 9], the sets of self-overlap lengths of v and w are $\{1, 2\}$ and $\{1\}$, respectively, and

$$B_v = B_w = \{00010,\ 0001,\ 00111\},$$

so the pair has property R although the sets of self-overlap lengths of v and w differ.

3. Localization

Throughout this section $n \geq 1$, and V and W are independent uniformly random words in $\{0,1\}^n$. We write $\mathbb{P}$ for the underlying probability and

$$p_n = \mathbb{P}\big((V, W) \text{ has property R}\big) = 4^{-n}|\mathcal{R}_n|.$$

Lemma 3.1. *Let $X_1, \dots, X_N$ be independent uniform bits, let $d \geq 1$ and $I \subseteq \{1, \dots, N-d\}$. Then*

$$\mathbb{P}\big(X_i = X_{i+d} \text{ for all } i \in I\big) = 2^{-|I|}.$$

Proof. Consider the graph on $\{1, \dots, N\}$ with the edges $\{i, i+d\}$, $i \in I$. A vertex is adjacent only to vertices at distance d, so each connected component is a set of consecutive elements of one residue class modulo d, joined in a path. The graph is therefore a forest with $|I|$ edges and $N - |I|$ components. The event in question says that the bits are constant on each component, and it has probability $2^{N-|I|}/2^N$. □

Lemma 3.2. *Let $s, t \in \{V, W\}$, where s and t may be the same random word.*

1. *For $1 \leq k \leq n-1$, $\mathbb{P}(\mathrm{suf}_k(s) = \mathrm{pre}_k(t)) = 2^{-k}$.*
2. *Let $1 \leq m \leq n$ and $1 \leq i, j \leq n-m+1$, with $i \neq j$ if s and t are the same word. Then $\mathbb{P}(s[i, i+m-1] = t[j, j+m-1]) = 2^{-m}$.*

Proof. If s and t are different words, the two compared blocks are independent and uniform, and both statements are clear. If $s = t$, the event in (1) is $s_i = s_{i+n-k}$ for $1 \leq i \leq k$, and Lemma 3.1 applies with $d = n-k$ and $I = \{1, \dots, k\}$. In (2) we may assume $i < j$; the event is $s_r = s_{r+d}$ for $i \leq r \leq i+m-1$, where $d = j - i$, and Lemma 3.1 applies with $I = \{i, \dots, i+m-1\} \subseteq \{1, \dots, n-d\}$. □

Now fix $K \geq 1$ and assume $n > 2K$; put $m = n - 2K$. For a pair (v, w) of words of length n write $a^{(1)} = v$ and $a^{(2)} = w$; the indices $1, 2$ distinguish the two positions in the pair even when $v = w$. We say that (v, w) satisfies condition (3.1) if, for all $c \in \{1, 2\}$ and $1 \leq k \leq K$,

$$\mathrm{suf}_k(a^{(c)}) = \mathrm{pre}_k(v) \iff \mathrm{suf}_k(a^{(c)}) = \mathrm{pre}_k(w). \tag{3.1}$$

We also consider the two exceptional conditions:

(X1) $\mathrm{suf}_k(a^{(c)}) = \mathrm{pre}_k(a^{(c')})$ for some $c, c' \in \{1,2\}$ and some k with $K < k < n$;

(X2) $\beta_{c,i} = \beta_{c',i'}$ for some $(c,i) \neq (c',i')$ in $\{1,2\} \times \{1,\dots,K+1\}$, where $\beta_{c,i} = a^{(c)}[i, i+m-1]$.

The blocks $\beta_{c,i}$ are well defined since $i + m - 1 \leq n - K$. Note that (X2) holds when $v = w$, since then $\beta_{1,1} = \beta_{2,1}$.

Lemma 3.3 (Localization)**.** *Let $K \geq 1$, $n > 2K$, and let (v,w) be a pair of words of length n satisfying neither* (X1) *nor* (X2)*. Then (v,w) has property R if and only if* (3.1) *holds.*

Proof. Since (X2) fails, $v \neq w$, and Proposition 2.2 applies. For $c \in \{1,2\}$ and $1 \leq k \leq K$ put $x_{c,k} = \mathrm{pre}_{n-k}(a^{(c)})$. Since (X1) fails, every overlap in (2.1) has length at most K, and

$$\begin{aligned} B_v &= \{x_{c,k} : c \in \{1,2\},\ 1 \leq k \leq K,\ \mathrm{suf}_k(a^{(c)}) = \mathrm{pre}_k(v)\}, \\ B_w &= \{x_{c,k} : c \in \{1,2\},\ 1 \leq k \leq K,\ \mathrm{suf}_k(a^{(c)}) = \mathrm{pre}_k(w)\}. \end{aligned} \tag{3.2}$$

We claim that for $(c,k) \neq (c',l)$ the word $x_{c,k}$ is not a suffix of $x_{c',l}$. Suppose it is. Comparing lengths gives $l \leq k$, and

$$a^{(c)}[1, n-k] = a^{(c')}[k-l+1,\ n-l].$$

Since $m = n - 2K \leq n - k$, comparing the first m letters gives $\beta_{c,1} = \beta_{c',k-l+1}$, where $1 \leq k - l + 1 \leq K$. The indices $(c,1)$ and $(c', k-l+1)$ are different: if $c = c'$, then $k \neq l$. This contradicts the failure of (X2) and proves the claim. In particular the words $x_{c,k}$ are pairwise distinct, so by (3.2) we have $x_{c,k} \in B_v$ if and only if $\mathrm{suf}_k(a^{(c)}) = \mathrm{pre}_k(v)$, and similarly for B_w.

By the claim, the only element of $B_v \cup B_w$ that is a suffix of an element x of this set is x itself. Hence (2.3) holds if and only if $B_v = B_w$, that is, if and only if $x_{c,k} \in B_v \Leftrightarrow x_{c,k} \in B_w$ for all c and k. By the preceding paragraph this is condition (3.1). □

Lemma 3.4. *For $K \geq 1$ and $n > 2K$,*

$$\begin{aligned} &\mathbb{P}\big((V,W) \text{ satisfies (X1)}\big) \leq 2^{2-K}, \\ &\mathbb{P}\big((V,W) \text{ satisfies (X2)}\big) \leq (K+1)(2K+1)\, 2^{-(n-2K)}. \end{aligned}$$

Proof. By Lemma 3.2(1) and a union bound over the four choices of (c,c'),

$$\mathbb{P}(\text{X1}) \leq 4 \sum_{k=K+1}^{n-1} 2^{-k} < 4 \cdot 2^{-K}.$$

There are $\binom{2K+2}{2} = (K+1)(2K+1)$ pairs of distinct indices in (X2), and by Lemma 3.2(2) each of the corresponding equalities has probability 2^{-m}. □

Proposition 3.5. *Let $K \geq 1$. The probability q_K that (V,W) satisfies* (3.1) *is the same for all $n > 2K$, and*

$$|p_n - q_K| \leq 2^{2-K} + (K+1)(2K+1)\, 2^{-(n-2K)} \qquad (n > 2K). \tag{3.3}$$

Proof. For $c \in \{1,2\}$ and $k \le K$, the words $\mathrm{suf}_k(a^{(c)})$ and $\mathrm{pre}_k(a^{(c)})$ are determined by $\mathrm{suf}_K(a^{(c)})$ and $\mathrm{pre}_K(a^{(c)})$, respectively, where $a^{(1)} = V$ and $a^{(2)} = W$. Hence (3.1) is a condition on the four blocks $\mathrm{pre}_K(V)$, $\mathrm{suf}_K(V)$, $\mathrm{pre}_K(W)$ and $\mathrm{suf}_K(W)$. Since $n > 2K$, these blocks occupy disjoint sets of positions, so they are independent and uniform in $\{0,1\}^K$, and the probability of (3.1) does not depend on n. By Lemma 3.3, the event that (V,W) has property R and the event (3.1) coincide outside $(\mathrm{X1}) \cup (\mathrm{X2})$. Hence $|p_n - q_K| \le \mathbb{P}(\mathrm{X1}) + \mathbb{P}(\mathrm{X2})$, and Lemma 3.4 gives (3.3). □

4. Words with prescribed boundary letters

In this section we bound p_n from below. Fix $L \ge 1$ and assume $n \ge 4L$. Let C_L be the event that each of V and W begins with 0^L and ends with 1^L. It fixes $2L$ distinct positions in each of the two independent words, so

$$\mathbb{P}(C_L) = 2^{-4L}, \tag{4.1}$$

and conditionally on C_L the remaining $2(n-2L)$ letters are independent uniform bits. We call the $4L$ prescribed positions *fixed* and the others *free*, and write $\mathbb{P}_L = \mathbb{P}(\,\cdot \mid C_L)$.

Lemma 4.1. *Let $s,t \in \{V,W\}$, let $1 \le k \le n-1$ and $d = n-k$, and let O be the event* $\mathrm{suf}_k(s) = \mathrm{pre}_k(t)$.

1. *If $k < 2L$, then $\mathbb{P}_L(O) = 0$.*
2. *If s and t are the same word and $k \ge 2L$, then $\mathbb{P}_L(O) \le 2^{-k}$.*
3. *If s and t are different words and $2L \le k \le n-L$, then $\mathbb{P}_L(O) = 2^{-k}$.*
4. *If s and t are different words and $n-L < k \le n-1$, then $\mathbb{P}_L(O) = 2^{-(n-2L+d)}$.*

Proof. The event O consists of the k equations $s_{d+j} = t_j$, $1 \le j \le k$.

(1) Let $j = \max(1, k-L+1)$. Then $j \le \min(k,L)$ because $k < 2L$, and $d+j \ge n-L+1$. Thus $s_{d+j} = 1$ and $t_j = 0$ on C_L, and O is impossible.

(2) Here O says that s is constant on each residue class modulo d in $\{1,\dots,n\}$. If $d < L$, the positions $1,\dots,L$ meet every residue class, so s would vanish identically, contradicting $s_n = 1$. Let $d \ge L$. Then the positions $1,\dots,L$ lie in distinct residue classes, and so do $n-L+1,\dots,n$. If one of the former and one of the latter are congruent modulo d, then O is impossible on C_L. Otherwise the $2L$ fixed positions of s lie in $2L$ distinct classes; then $d \ge 2L$, and exactly 2^{d-2L} of the 2^{n-2L} words with the prescribed boundary are constant on all classes. Hence $\mathbb{P}_L(O) = 2^{-k}$ in this case.

(3) and (4). Now s and t are different words, and the equations pair the positions $d+1,\dots,n$ of s with the positions $1,\dots,k$ of t, each position being used once. Different equations therefore involve disjoint sets of free letters. An equation with at least one free letter holds with conditional probability $1/2$, independently of the others, and an equation between two fixed letters holds with probability 1 or 0. We list the equations between fixed letters, using $k \ge 2L$ and $n \ge 4L$.

- A final letter of s (position $d+j \geq n-L+1$, that is, $j \geq k-L+1$) against an initial letter of t ($j \leq L$): impossible, since $k-L+1 > L$.
- An initial letter of s ($d+j \leq L$) against an initial letter of t: this happens exactly for $1 \leq j \leq L-d$, and both letters are 0.
- A final letter of s against a final letter of t ($j \geq n-L+1$): this happens exactly for $n-L+1 \leq j \leq k$, that is, for $L-d$ values of j if $d < L$, and both letters are 1.
- An initial letter of s against a final letter of t: impossible, since $L-d < n-L+1$.

If $k \leq n-L$, then $d \geq L$ and no equation joins two fixed letters, so $\mathbb{P}_L(O) = 2^{-k}$. If $k > n-L$, then $1 \leq d < L$, exactly $2(L-d)$ equations join two equal fixed letters, and each of the remaining $k-2(L-d) = n-2L+d$ equations contains a free letter. This proves (4). □

Proposition 4.2. *For $L \geq 1$ and $n \geq 4L$,*

$$p_n \geq 2^{-4L}\big(1 - 2^{3-2L} - 2^{1-(n-2L)}\big).$$

Proof. Let G be the event that (s,t) has no overlap of any length $1 \leq k \leq n-1$, for each of the four choices of $s,t \in \{V,W\}$. If a pair (v,w) lies in G and $v \neq w$, then $B_v = B_w = \emptyset$ and (v,w) has property R by Proposition 2.2; if $v = w$, it has property R by Remark 2.3. Hence $p_n \geq \mathbb{P}(C_L)\,\mathbb{P}_L(G)$.

By Lemma 4.1, for each of the two choices with $s = t$,

$$\sum_{k=1}^{n-1} \mathbb{P}_L\big(\mathrm{suf}_k(s) = \mathrm{pre}_k(s)\big) \leq \sum_{k=2L}^{n-1} 2^{-k} < 2^{1-2L},$$

and for each of the two choices with $s \neq t$,

$$\sum_{k=1}^{n-1} \mathbb{P}_L\big(\mathrm{suf}_k(s) = \mathrm{pre}_k(t)\big) = \sum_{k=2L}^{n-L} 2^{-k} + \sum_{d=1}^{L-1} 2^{-(n-2L+d)} < 2^{1-2L} + 2^{-(n-2L)},$$

where the second sum is empty if $L = 1$. By the union bound, $1 - \mathbb{P}_L(G) < 4 \cdot 2^{1-2L} + 2 \cdot 2^{-(n-2L)}$, and (4.1) completes the proof. □

Corollary 4.3. *For every $n \geq 8$, $p_n \geq 2^{-9} - 2^{-(n+3)}$.*

Proof. For $L = 2$ the bound of Proposition 4.2 equals $2^{-8}\big(1 - 2^{-1} - 2^{5-n}\big) = 2^{-9} - 2^{-(n+3)}$. □

Remark 4.4. The case $L = 2$ is the construction of words beginning with 00 and ending with 11 described in the remark following Theorem 3.6 of Bóna, Maga and Richey [1, p. 10]. The point of Lemma 4.1(4) is that for overlaps of length close to n the boundary letters of s and t are matched against each other, so that the conditional probability $2^{-(n-2L+d)}$ is larger than 2^{-k}; the sum of these terms is still at most $2^{-(n-2L)}$. Among the values of L, the choice $L = 2$ gives the best bound in Proposition 4.2: for $L = 1$ the bound is negative, and for $L \geq 3$ it is less than 2^{-12}.

Remark 4.5. The lower bound 1/512 is far below the observed proportions. Table 1 was computed by enumerating all ordered pairs of length n and testing condition (2.3) of Proposition 2.2, with the diagonal pairs included as in Remark 2.3; the C program and its output are provided as supplementary files. These values are not used in the proofs. For comparison, Drexel, Peng and Richey [2, p. 8] report the approximate value 0.15753 from an exact computer search with $n = 15$; our enumeration gives 0.15753 for $n = 10$ and 0.15342 for $n = 15$.

TABLE 1. The number of ordered pairs of length n with property R, and the proportion $p_n = 4^{-n}|\mathcal{R}_n|$.

n	$\lvert\mathcal{R}_n\rvert$	p_n	n	$\lvert\mathcal{R}_n\rvert$	p_n
1	4	1.00000	9	41880	0.15976
2	8	0.50000	10	165180	0.15753
3	16	0.25000	11	653600	0.15583
4	56	0.21875	12	2598120	0.15486
5	188	0.18359	13	10353508	0.15428
6	716	0.17480	14	41286284	0.15380
7	2732	0.16675	15	164729736	0.15342
8	10636	0.16229			

5. Proofs of the main theorems

Proof of Theorem 1.1. Let $\varepsilon > 0$. Choose $K \geq 1$ with $2^{3-K} < \varepsilon/2$, and then $N > 2K$ with $2(K+1)(2K+1)\,2^{-(N-2K)} < \varepsilon/2$. For $n, r \geq N$, Proposition 3.5 applied to n and to r with the same K gives

$$\begin{aligned}|p_n - p_r| &\leq |p_n - q_K| + |q_K - p_r| \\ &\leq 2^{3-K} + (K+1)(2K+1)\big(2^{-(n-2K)} + 2^{-(r-2K)}\big) < \varepsilon.\end{aligned}$$

Thus $(p_n)_{n\geq 1}$ is a Cauchy sequence and converges to some $\rho \in [0,1]$. By Corollary 4.3, $\rho \geq 2^{-9} = 1/512$.

By (2.4), $4^{-n}|\mathcal{R}_n^{\neq}| = p_n - 2^{-n}$ and $|\mathcal{R}_n^{\neq}|/(4^n - 2^n) = (p_n - 2^{-n})/(1 - 2^{-n})$, and both tend to ρ. □

Proof of Theorem 1.2. Letting $n \to \infty$ in (3.3) gives $|\rho - q_K| \leq 2^{2-K}$. Now let $n \geq 3$ and $K = \lfloor n/3 \rfloor \geq 1$. Then $n - 2K \geq K$, so $n > 2K$, and by (3.3)

$$|p_n - \rho| \leq |p_n - q_K| + |q_K - \rho| \leq 2^{3-K} + (K+1)(2K+1)\,2^{-K}.$$

Since $K > n/3 - 1$, the right-hand side is at most $2\big((K+1)(2K+1) + 8\big)2^{-n/3} = O(n^2 2^{-n/3})$. □

Remark 5.1. By Theorem 1.2, ρ is computable: q_K is determined by an enumeration of the 2^{4K} possible quadruples of boundary blocks, and it approximates ρ within 2^{2-K}. This is not an efficient method of computing ρ

to high precision, and the rate in Theorem 1.2 has not been optimized. We do not know a closed form for ρ.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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